

National University of Singapore

CS1101S — Programming Methodology

AY2023/2024 Semester 1

Final Assessment

Time allowed: 2 hours

ANSWER SHEET

STUDENT NUMBER									
A									
U	☐	0	0	0	0	0	0	0	A N
A	☒	1	1	1	1	1	1	1	B R
HT	☐	2	2	2	2	2	2	2	E U
NT	☐	3	3	3	3	3	3	3	H W
		4	4	4	4	4	4	4	J X
		5	5	5	5	5	5	5	L Y
		6	6	6	6	6	6	6	M
		7	7	7	7	7	7	7	
		8	8	8	8	8	8	8	
		9	9	9	9	9	9	9	

(1) [3 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(2) [2 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(3) [3 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(4) [3 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(5) [4 marks]

```
function power(x, n) {
    return 
    ? 
    : 
}
```

(6) [4 marks]

```
function app_count(f) {
    let count = 0;
    function wrapper(x) {
        
    }
    
}
```

(7) [5 marks]

```

function fancy_sum(xs) {
  return accumulate(
    [
    ],
    map(
      filter(
        xs
      )
    )
  );
}

```

(8) [4 marks]

```

const S = list(1, 2, 3, 4, 5);

set_head(
  [
  ]
);

set_head(
  [
  ]
);

```

(9) [3 marks]

```

function flatten_SCL(S) {
  return filter(
    [
    ]
  );
}

```

(10) [5 marks]

```

function skip_to(S, idx) {
  function helper(scl, idx) {
    [
    ]
  }
  set_head(
    [
    ]
  );
}

```

(11) [5 marks]

```
function diameter(S) {
    if (is_null(S)) {
```

```
    } else if ( ! is_pair(head(S)) ) {
```

```
    } else {
```

```
    }
```

```
}
```

(12) [3 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(13) [3 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(14) [3 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(15) [3 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(16) [3 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(17) [4 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(18) [4 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(19) [2 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(20) [2 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H
(21) [2 marks]
☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(22) [3 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

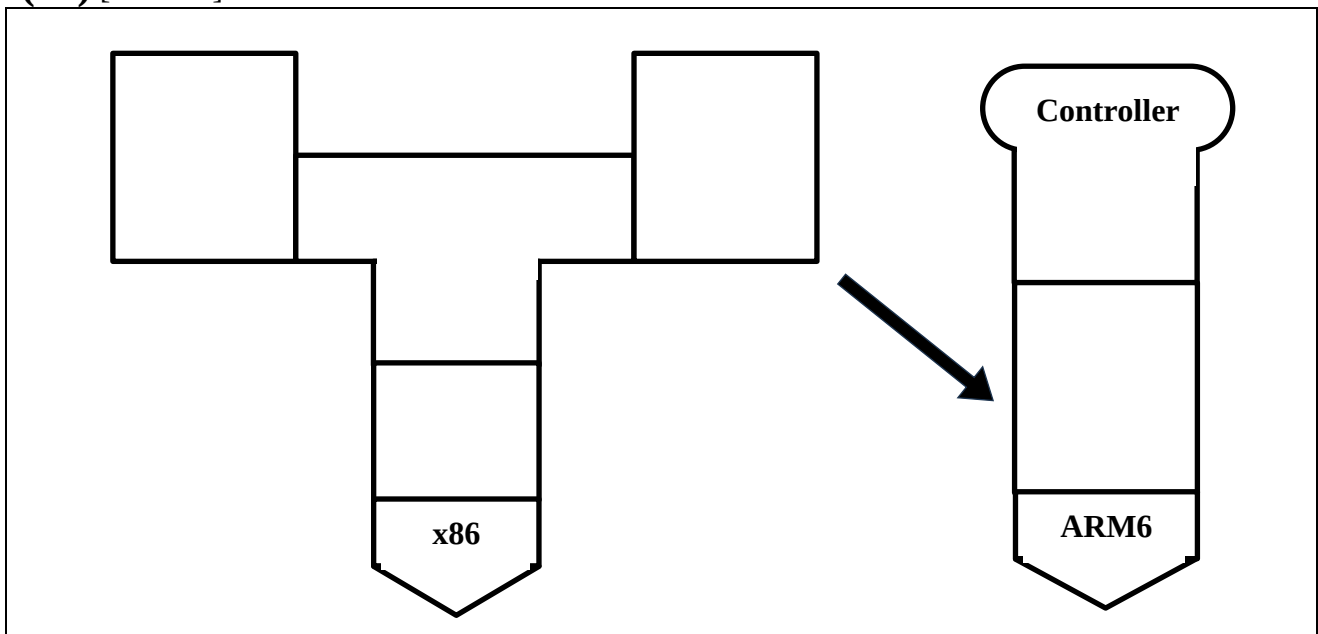
(23) [3 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(24) [5 marks]

```
function repeat(S, k) {
  return is_null(S)
    ? null
    : stream_append(
      ,
    );
}
```

(25) [8 marks]



(26) [3 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(27) [4 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

(28) [4 marks]

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

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